

BOTSON HOUSE PRICE PREDICTION USING ML DOCUMENTATION

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INTRODUCTION

Description Boston House Price Prediction is a machine learning project used to estimate house prices based on housing and neighbourhood characteristics.

Use Case

Real estate price estimation

Property valuation

Investment planning

Market analysis

DATA PREPROCESSING

Description Data preprocessing prepares the dataset for machine learning by cleaning and transforming the data.

Key Activities

Handling missing values

Removing outliers

Feature scaling

Data normalization

Output A clean dataset ready for training machine learning models.

Exploratory Data Analysis (EDA)

Description EDA helps understand the dataset and identify relationships between features and house prices.

Use Case

Finding important features

Detecting trends

Understanding data distribution

Output Graphs and visualizations showing feature relationships and correlations.

Linear Regression

Description Linear Regression predicts house prices by finding a linear relationship between input features and the target variable.

Advantages

Simple implementation

Easy interpretation

Fast training

Output Predicted house prices based on linear relationships.

Decision Tree Regression

Use Case

Description Decision Tree Regression splits data into branches and predicts values based on decision rules.

Advantages

Handles non-linear data

Easy visualization

Better interpretability

Output House price predictions using tree-based decision paths.

Random Forest Regression

Description Random Forest combines multiple decision trees to improve prediction accuracy.

Advantages

Reduces overfitting

High accuracy

Robust predictions

Output More accurate house price predictions compared to a single decision tree.

Gradient Boosting Regression

Description Gradient Boosting builds models sequentially and improves predictions by correcting previous errors.

Advantages

High performance

Better accuracy

Handles complex relationships

Output Optimized house price predictions.

Evaluation Metrics

Description Evaluation metrics measure how well the model predicts house prices.

Metrics

Mean Squared Error (MSE)

Root Mean Squared Error (RMSE)

R^2 Score

Output Performance scores used to compare different models.

Results

Description The performance of all machine learning models is compared.

Output

Predicted house prices

Accuracy scores

Best-performing model selected

Conclusion

Description Machine learning models successfully predict house prices using housing features.

Outcome

Improved prediction accuracy

Better decision-making support

Practical real-estate application